MA'ADEN معادن

MA'ADEN ENGINEERING MANUAL

Anchor Bolts

MD-101-SMEM-EG-CV-SPC-0002

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1. INTRODUCTION

1.1 Scope

This document provides the details and requirements for anchor bolt fabrication and installation.

This document is intended to be used in conjunction with the applicable documents listed herein for use in all projects of Ma'aden.

1.2 Purpose

The purpose of this specification is to provide details and requirements for anchor bolt fabricators and installers.

2. KEY TERMS AND DEFINITIONS

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
ACI	American Concrete Institute.
ASTM	American Society for Testing and Materials.
SBC	Saudi Building Code.

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards or codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

The work shall conform to the following applicable codes and standards.

3.2.1 Local Codes and Standards

3.2.1.1 Saudi Building Code

SBC 303	Structural – Soil and Foundation
SBC 304	Structural – Concrete Structure
SBC 306	Structural – Steel Structures.

3.2.2 International Codes and Standards

3.2.2.1 American Society for Testing and Materials (ASTM)

ASTM A6/6M	Rolled Structural Steel Bars, Plate Shapes and Sheet Piling.
ASTM A36/A36M	Standard Specification for Carbon Structural Steel.
ASTM A153/A153M	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
ASTM A193/A193M	Standard Specification for Alloy Steel and Stainless Steel Bolting Material for High Temperature Service.
ASTM A194/A194M	Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High Temperature Service.
ASTM A307	Specification for Carbon Steel Bolts and Studs, 60000 psi Tensile Strength.
ASTM F436	Standard Specification for Hardened Steel Washers.
ASTM A563	Standard Specification for Carbon and Alloy Steel Nuts.
ASTM F436	Specification for Hardened Steel Washers.
ASTM F1554	Standard Specification for Anchor Bolts, Steel, 36, 55, 105 ksi Yield Strength.
ASTM F2329	Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts and Special Threaded Fasteners.

3.3 Ma'aden Standards

MD-101-SMEM-EG-ST-DST-0001-05	Structural Standard Drawing – Anchor Bolt Details
MD-101-SMEM-EG-ST-SPC-0001	Steelwork Fabrication.
MD-101-SMEM-EG-ST-SPC-0002	Steelwork Erection.
MD-101-SMEM-EG-ST-SPC-0003	Grout.
MD-101-SMEM-EG-ST-SPC-0005	Concrete Works.
MD-101-SMEM-EG-ST-SPC-0009	Protective Coating for Steelwork.
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

4. MANUFACTURE

4.1 Material

Generally, anchor bolts shall conform to ASTM F1554 Section 6.6.1 and Table 3.

Nuts shall conform to ASTM A563, Grade A, heavy hex with UNC-2B threads. Washers shall be ASTM F436.

Anchor plate shall conform to ASTM A36.

4.2 Fabrication

Unless otherwise specified, all bolts (total bolt length), nuts and washers shall be hot-dip galvanized after fabrication in accordance with ASTM A153, Class C.

Excess galvanizing material shall be removed from the threaded portions of the bolts by the use of a centrifuge or by mechanical chasing of the threads.

The fit of nuts on threads of anchor bolts shall be verified before shipment.

4.3 Tolerances

The tolerances on overall lengths of any anchor bolt shall be +0 mm – 25 mm, hooks shall have a tolerance ±6 mm on overall dimensions after bending.

The tolerance in setting out bolts in group with a specified template shall comply with ACI 318.

5. INSTALLATION

5.1 General

Anchor bolts details type A, B and C are shown on the Standard Drawing MD-101-SMEM-EG-ST-DST-0001-05.

Type A bolt assemblies shall consist of anchor bolts with tack-welded plate. Type B and C assemblies shall consist of anchor bolts with cog or U shape bars only, with a certain amount of embedment depth.

Unless noted otherwise on the design drawings, anchor bolt sleeves for stationary equipment and structural base plates shall be filled with grout after the column or equipment is in place and the anchor bolts are aligned. Water or other loose particles shall not be allowed to collect in the sleeve before the sleeve is grouted. Grouting procedure shall conform to the Standard Specification MD-101-SMEM-EG-ST-SPC-0003, Grout.

For machines and equipment anchor bolts, machine/equipment vendor shall specify anchor bolts size and spacing to the Employer for incorporation with the machine/equipment support design.

Unless noted otherwise on the design drawings, anchor bolts sleeves for machinery shall be filled with non-bonding moldable material before grouting.

Anchor bolt threads shall be covered with duct tape or other suitable means to keep them clean and to prevent any damage that might occur during the preparation of the foundation for grouting and the actual grouting of the foundation.

Unless a torque value is specified on design drawings or at the locations of slide plates, anchor bolts shall be tightened to a snug condition. At slide plate locations, nuts shall be backed off a half turn and a double nuts shall be added.

All anchor bolts subjected to sulfurous gases, mists and fumes as well as high humidity shall be protected against corrosion by painting to the same paint system as structural steel after the structural steel is erected or the associated equipment is installed. The Contractor shall confirm surface preparation and primer type are acceptable for chosen paint manufacturer. All anchor bolts subjected to sulfurous gases, mists and fumes as well as high humidity shall be protected against corrosion with the same protective coating that applied to the structural steel, which is specified in the standard specification MD-101-SMEM-EG-ST-SPC-0008, Protective Coating for Steelwork.

5.2 Sleeves

Anchors should be installed with sleeves when small movement of the bolt is desired after the bolt is set in concrete. The two most common examples follow:

a) When precise alignment of anchors is required during installation of structural columns or equipment. In this situation, the sleeve should be filled with grout after installation is complete.

b) When anchors will be pre-tensioned to maintain the bolt under continuous tensile stresses during load reversal. Pre-tensioning requires the bolt surface to be free; therefore, the top of these sleeves should be sealed or the sleeve should be filled with elastomeric material to prevent grout or water from filling the sleeve.

Refer to API RP 686 for use of sleeves with anchor bolts in machinery foundations.

6. QUALITY ASSURANCE

6.1 General

A Quality Assurance/Inspection Plan shall be provided, detailing procedures to be employed and the responsible personnel, to ensure that the requirements of this Specification and the applicable codes and standards are met as specified in MD-101-SMPM-PM-QA-PEI-0001, Ma'aden Contractor Quality Requirements Project Execution Instruction.

The Employer shall have the authority to reject materials, procedures or work which, in its opinion fail to meet the requirements of the Specification, the Contract, or Documents referred to in this Specification.

6.2 Inspection Requirement

The following inspections shall be carried out by the Contractor as a minimum.

6.2.1 Material

Inspection procedures shall be developed to ensure that materials meet the requirements of Steelwork Fabrication Specification, prior to installation.

6.2.2 Identification Requirement

Group of identical bolts shall be securely tied together for shipment and tagged according to concrete drawing number followed by schedule item number.

Tags shall consist of stamped aluminium or zinc strips securely wired to the bolts.

6.2.3 Bolt Protection

Visual inspection of surface treatment shall be carried out. Any defects of the bolts shall be reported and to be replaced. Installed anchor bolts shall be cover with a plastic cap.

6.2.4 Field Welding

Welding inspection and testing frequency for field welding shall be the same as specified for shop welding in Steel Fabrication Specification.

The Contractor shall specify in the inspection testing plan the methodology to be employed in the sampling and selection of the welder and the welds to be inspected.

6.2.5 Bolting

Anchor bolts shall be tightened to a snug condition unless noted otherwise on the design drawings.